OVER A BEZOUT DOMAIN ANY UNIMODULAR ROW OF LENGTH N IS THE FIRST ROW OF AN INVERTIBLE MATRIX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

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Editorial note. The statement writes 'whose first row is f_1 \ldots f_n' using the ellipsis where a comma-separated list is meant; that is the author's text and is reproduced unchanged. The single reference inside the counted proof span is recorded as a non-load-bearing navigation pointer.

Allowed standard prerequisites (editorial).

- Bezout domains: finitely generated ideals are principal. used at proof lines 37937 and 37940-37941. Allows the ideal generated by the first n - 1 entries to be written as (f) and produces the relation af - b f_n = 1.
- Invertibility of a matrix with determinant one. used at proof line 37962. Certifies that the constructed matrix is invertible.

Source reference. Original lemma 37847: sources/stacks-more-algebra/pinned-original-source.tex, source lines 37847–37847. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a Bézout domain. Let $n \geq 1$ and $f_1, \ldots, f_n \in R$ generate the unit ideal. There exists an invertible $n \times n$ matrix in R whose first row is $f_1 \ldots f_n$.*

Proof. This follows from Lemma 37847 but we can also prove it directly as follows. By induction on n. The result holds for $n = 1$. Assume $n > 1$. We may assume $f_1 \neq 0$ after renumbering. Choose $f \in R$ such that $(f) = (f_1, \ldots, f_{n-1})$. Let A be an $(n-1) \times (n-1)$ matrix whose first row is $f_1/f, \ldots, f_{n-1}/f$. Choose $a, b \in R$ such that $af - bf_n = 1$ which is possible because $1 \in (f_1, \ldots, f_n) = (f, f_n)$. Then a solution is the matrix

$$\begin{pmatrix} f & 0 & \ldots & 0 & f_n \\ 0 & 1 & \ldots & 0 & 0 \\ & & \ldots & & \\ 0 & 0 & \ldots & 1 & 0 \\ b & 0 & \ldots & 0 & a \end{pmatrix} \begin{pmatrix} & & & 0 \\ & A & & \\ & & & 0 \\ 0 & \ldots & 0 & 1 \end{pmatrix}$$

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Observe that the left matrix is invertible because it has determinant 1. □